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Influence of the pre-ionization background and simulation of the optical emission of a streamer discharge in preheated air at atmospheric pressure between two point electrodes

A Bourdon¹, Z Bonaventura^{1,2} and S Celestin³

¹ Ecole Centrale Paris, EM2C Laboratory, UPR CNRS 288, Grande voie des vignes, 92295 Châtenay-Malabry Cedex, France

² Masaryk University, Faculty of Science, Department of Physical Electronics, Kotlářská 2, 611 37 Brno, Czech Republic

³ Communications and Space Sciences Laboratory, Department of Electrical Engineering, The Pennsylvania State University, University Park, PA 16802, USA

E-mail: anne.bourdon@em2c.ecp.fr

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Abstract

This paper presents simulations of positive and negative streamers propagating between two point electrodes in preheated air at atmospheric pressure. As many discharges have occurred before the simulated one, seed charges are taken into account in the interelectrode gap. First, for a pre-ionization background of 10^9 cm^{-3} , we have studied the influence of the data set used for transport parameters and reaction rates for air on the simulation results. We have compared results obtained in 1997 using input parameters from Morrow and Lowke and from Kulikovsky. Deviations as large as 20% of streamer characteristics (i.e. electric field in the streamer head and body, streamer velocity, streamer radius, streamer electron density) have been observed for this point-to-point configuration. Second, we have studied the influence of the pulsed voltage frequency on the discharge structure. For the studied discharge regime, a change in the applied voltage frequency corresponds to a change in the pre-ionization background. In this work, we have considered a wide range of pre-ionization values from 10^4 and up to 10^9 cm^{-3} . We have noted that the value of the pre-ionization background has a small influence on the electron density, electric field and location of the negative streamer head. Conversely, it has a significant influence on the positive streamer characteristics. Finally, we have compared instantaneous and time-averaged optical emissions of the three band systems of N_2 and N_2^+ (1PN_2 , 2PN_2 and 1NN_2^+) during the discharge propagation. We have shown that the emission of the 2PN_2 is the strongest of the three bands, in agreement with experimental observations. It is interesting to note that even with a short time averaging of a few nanoseconds, which corresponds to currently used instruments, the structure of the time-averaged emission of the 2PN_2 is different from the instantaneous one and shows negative and positive streamers with smaller radial expansions and more diffuse streamer heads.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

In recent years, plasma discharges at atmospheric pressure have gained popularity due to their ability to produce active species in well-controlled environments and at very low energy cost. Their potential has been demonstrated for a wide range of applications including plasma-assisted combustion, biomedical treatment and decontamination and thin film coatings (van Veldhuizen 2000, Fridman *et al* 2005). In air at atmospheric pressure, discharges usually take the form of thin plasma filaments, also called streamers. It is interesting to note that similar filamentary structures are observed in sprites, large-scale discharges which appear at low pressure in the altitude range $\sim 40\text{--}90$ km above large thunderstorms (Sentman *et al* 1995). A review of current knowledge on sprites emphasizing similarity with laboratory discharges is given in Pasko (2007).

Due to the wide variety of applications, studies on discharges at atmospheric pressure are carried out for a large spectrum of reactive gases. However, a significant number of applications are using air as carrier gas and then numerous experimental and simulation studies have been done to understand discharge characteristics in air. In particular, this is the case for the plasma-assisted combustion of lean premixed flames. In this application, the concentration of the fuel is very low in the fuel/air mixture and then main active species are created by the discharge in air. These active species as atomic oxygen, even in low concentrations have a significant influence on the combustion chemistry (Starikovskaia 2006). Therefore, to understand and optimize the effect of the discharge on the lean flame ignition or stabilization, it is necessary to carry out detailed studies in air.

Then for example, in parallel with experimental studies on plasma-assisted combustion based on the use of a nanosecond repetitively pulsed (NRP) discharge between two pins (Pilla *et al* 2006), a thorough experimental study was carried out on the discharge in preheated air at atmospheric pressure with the same generator and the same geometry (Pai *et al* 2008, Pai 2008). It is interesting to note that the cumulative effect of repeated pulsing achieves steady-state behaviour. Thus, even though they are transient, the NRP discharges have a visual resemblance with dc discharge regimes. For an air temperature of 1000 K, as the applied voltage increases, three different discharge regimes are observed. At low voltage, the 'C' regime, like a corona discharge emits light only near the point electrodes. For voltages higher than 6 kV, the 'D' regime is observed. This regime has an emission which fills the gap in a diffuse manner. Finally, for voltages higher than 7 kV, an 'F' regime is observed. This regime is reminiscent of a spark discharge with an intense emission. The measurements show that the 'D' regime has low levels of emission, gas heating and electrical conduction current, indicating that it is non-thermal. Furthermore, this regime develops through a positive streamer propagating between both electrodes followed by a return wave in the opposite direction (Pai 2008). In contrast to the non-thermal 'D' regime, the thermal 'F' regime emits strongly, heats the gas by several thousand degrees Kelvin and

has tens of amperes of conduction current. Further details on the experimental results may be found in Pai (2008). In parallel to this thorough experimental study in air, we have developed a 2D streamer code to simulate the discharge in a point-to-point geometry (Celestin 2008). The objective of these complementary experimental and numerical studies is to better understand the transitions between the different discharge regimes and finally to optimize the effect of the discharge on a lean flame. First simulations of the dynamics of a discharge in preheated air (at 1000 K) in the point-to-point geometry (with a 5 mm interelectrode gap) have been presented in Celestin *et al* (2009) for a constant applied voltage of 7 kV. The ghost fluid method has been used to solve Poisson's equation on a rectilinear grid by taking into account the exact shapes of the electrodes in the simulation domain (Celestin *et al* 2009). We have shown that positive and negative streamers propagate from the anode and the cathode, respectively. Then, after the interaction of both discharges, the positive streamer propagates very rapidly towards the cathode in the volume pre-ionized by the negative streamer. This structure of the discharge is in qualitative agreement with the experiment and corresponds to the 'D' regime. In Celestin *et al* (2009), to take into account the fact that the discharge occurs at atmospheric pressure in preheated air at $T = 1000$ K, we have only changed the value of the total density $N = N_0 T_0 / T$ where N_0 is the density of air at ground pressure and ambient temperature ($T_0 = 273$ K). This decrease in the total density by a factor of 3, increases by the same factor the local reduced electric field E/N and then has a direct impact on transport parameters and reaction rates for air which are assumed to be functions of E/N . However, it is important to note that at $T = 1000$ K, the composition of air is slightly different from the one at ambient temperature and this may have also an impact on transport parameters and source terms. In fact, it is rather difficult to estimate accurately the influence of this effect as, even at ambient temperature, the coefficients used in streamer simulations are only approximations and significant quantitative differences exist between the data sets used in the literature at atmospheric pressure (e.g. Morrow and Lowke (1997), Kulikovskiy (1997), Liu and Pasko (2004)). Then in this work, as a first step of a more detailed study on the influence of the change in the air composition on the discharge characteristics, we study the influence of the data set used for transport parameters and source terms on the streamer dynamics.

In Celestin *et al* (2009), we have simulated a single discharge in air occurring during one of the 10 ns long voltage pulses. As many discharges have occurred before the simulated one, we have estimated that the density of seed charges in the interelectrode gap is on the order of 10^9 cm^{-3} for the 10–30 kHz voltage frequency range. In this work, we discuss the influence of the pulsed voltage frequency on the discharge structure. Experimentally, for an applied voltage up to 7 kV and for frequencies up to 10 kHz, the discharge is either in the 'C' or in the 'D' regime and therefore the gas heating is negligible. Then, the change in the pulsed voltage frequency only impacts the pre-ionization background. In Pancheshnyi *et al* (2001) and

Pancheshnyi (2005), the influence of seed charges on the characteristics of a positive streamer in the point-to-plane geometry has been shown. In this work, we study the influence of seed charges, that is of the frequency of the applied NRP voltage, on positive and negative streamers in a point-to-point geometry. To determine accurately the spatial distribution of seed charges at the beginning of the studied discharge, it would be necessary to simulate the 2D dynamics of the repetitively pulsed discharge, which is beyond the scope of this study. Then, in this work, as in Celestin *et al* (2009), we have assumed that, as a first approximation, seed charges are distributed uniformly in the computational domain.

Furthermore, in order to compare more easily simulation results with experiments we calculate optical emissions of the discharge. Emissions of streamers mainly consist of band systems from N_2 and N_2^+ : the first positive [$1PN_2$, $N_2(B^3\Pi_g) \rightarrow N_2(A^3\Sigma_u^+)$] and second positive [$2PN_2$, $N_2(C^3\Pi_u) \rightarrow N_2(B^3\Pi_g)$] band systems of N_2 , and the first negative band system of N_2^+ [$1NN_2^+$, $N_2^+(B^2\Sigma_u^+) \rightarrow N_2^+(X^2\Sigma_g^+)$]. In laboratory experiments, the emission of streamers at atmospheric pressure and room temperature is dominated by the blue emission of the $2PN_2$ (Shcherbakov and Sigmond 2007). In Liu *et al* (2008), the instantaneous optical emissions of the three systems ($1PN_2$, $2PN_2$ and $1NN_2^+$) for a positive streamer propagating in a non-uniform field created by a high voltage sphere has been simulated at atmospheric pressure. These modelling results show that the emission of the $2PN_2$ is stronger than $1PN_2$ and $1NN_2^+$ emissions. It is interesting to note that for low-pressure sprite streamers, the recorded emissions are dominated by red emissions of the $1PN_2$, which is in agreement with modelling results (Liu *et al* 2009). It is important to mention that in experiments, the measured emission at a given time is not instantaneous, but time averaged. In sprite experiments, recent high-speed video recordings have been carried out with a 50 μs temporal resolution. The influence of time integration on the optical emissions of sprites has been recently studied in Liu *et al* (2009). In laboratory experiments, high temporal resolution (~ 1 ns) images have been obtained recently for streamer development and reveal localized and bright streamer heads. These high-resolution images are really different from images obtained with longer timescales (a few microseconds) which show continuous luminous channels (Briels *et al* 2006). In Pancheshnyi *et al* (2005), the optical emission of the $2PN_2$ of a streamer discharge in the point-to-plane geometry has been calculated for accumulation times of 0.1, 1.0 and 5 ns at 500 Torr. In the experiments on the NRP discharge (Pai 2008), the time resolution of the camera used is 2 ns. In this work, we calculate instantaneous and time-averaged intensities of the three systems ($1PN_2$, $2PN_2$ and $1NN_2^+$) for a discharge in the point-to-point geometry at 760 Torr in preheated air at 1000 K. The influence of the time integration on the interpretation of the results is discussed.

2. Model formulation

2.1. Streamer equations

The most common and effective model to study the dynamics of streamers is based on the following drift–diffusion equations for electrons and ions coupled with Poisson’s equation (e.g. Kulikovsky (1997)):

$$\frac{\partial n_e}{\partial t} + \vec{\nabla} \cdot n_e \vec{v}_e - \vec{\nabla} \cdot (D_e \vec{\nabla} n_e) = S_{ph} + S_e^+ - S_e^-, \quad (1)$$

$$\frac{\partial n_p}{\partial t} = S_{ph} + S_p^+ - S_p^-, \quad (2)$$

$$\frac{\partial n_n}{\partial t} = S_n^+ - S_n^-, \quad (3)$$

$$\nabla^2 V = -\frac{q_e}{\epsilon_0}(n_p - n_n - n_e), \quad (4)$$

where subscripts ‘e’, ‘p’ and ‘n’ refer to electrons, positive and negative ions, respectively, n_i is the number density of species i , V is the potential, $\vec{v}_e = -\mu_e \vec{E}$ ($\vec{E}$ being the electric field) is the drift velocity of electrons, D_e and μ_e are the diffusion coefficient and the absolute value of mobility of electrons, respectively, q_e is the absolute value of electron charge and ϵ_0 is the permittivity of free space. On timescales of interest for this work, we have checked that the influence of ion motions on the results presented in this paper is negligible and then, to simplify the calculations, ions are assumed to be motionless. The S^+ and S^- terms stand for the rates of production and loss of charged particles. The S_{ph} term is the rate of electron–ion pair production due to photoionization in a gas volume.

All the coefficients of the model are assumed to be functions of the local reduced electric field E/N , where E is the electric field magnitude and N is the air neutral density. For test studies presented in this paper all transport parameters and reaction rates for air are taken from Morrow and Lowke (1997). For the photoionization source term, the three-group SP_3 model is used (Bourdon *et al* 2007) with boundary conditions given in Liu *et al* (2007). In this paper axisymmetric streamers are studied and thus cylindrical coordinates (x, r) are used with the x -axis as axis of the discharge. The charged species transport equations are solved using a modified Scharfetter–Gummel algorithm (Kulikovsky 1995). In this work, as in Celestin *et al* (2009) we have taken into account simplified boundary conditions: near the anode and cathode surfaces, gradients of electron density are assumed to be zero. The finite difference form of Poisson’s equation is solved using the D03EBF module of the NAG Fortran library (<http://www.nag.co.uk>) with the same boundary conditions as in Celestin *et al* (2009). For the photoionization source term, we use the SuperLU solver (Demmel *et al* 1999a,b) (<http://crd.lbl.gov/~xiaoye/SuperLU/>). To define the time step of the simulation, we use the same approach as in Bourdon *et al* (2007). The point-to-point geometry is the same as the one used in Celestin *et al* (2009): two hyperboloid electrodes with a radius of curvature of $\sim 300 \mu m$ separated by

5 mm. In this work, we have used a $2 \times 0.5 \text{ cm}^2$ computational domain discretized on a fixed rectilinear grid with $n_x \times n_r = 1510 \times 347$ cells. The grid is Cartesian, with a fixed cell size of $5 \mu\text{m}$ in the region where the streamer propagates. Beyond this region the grid expands following a geometric progression. The tip of the anode is located at $x = 1.25 \text{ cm}$ and the tip of the grounded cathode is located at $x = 0.75 \text{ cm}$.

2.2. Optical emissions of the discharge

To calculate optical emissions of the 1PN_2 , 2PN_2 and 1NN_2^+ bands, we use a model similar to the one given in Pasko *et al* (1997). The optical emission intensity of each band at a given time t is given by

$$I_k = 10^{-6} \int_{\mathcal{L}} A_k n_k d\mathcal{L}, \quad (5)$$

where the intensity I_k is in Rayleighs, $n_k \text{ (cm}^{-3}\text{)}$ is the population of excited state k , $A_k \text{ (s}^{-1}\text{)}$ is the radiation transition rate. The integration is performed along the optical path $\mathcal{L}$. In this paper, as well as in Pasko *et al* (1997) and Liu and Pasko (2004), the effects of radiative transfer between the source of the emission and the observer are not taken into account. The population of excited states is governed by

$$\frac{\partial n_k}{\partial t} = -\frac{n_k}{\tau_k} + \sum_m n_m A_m + \nu_k n_e, \quad (6)$$

where $\tau_k = [A_k + \alpha_{\text{N}_2}^k N_{\text{N}_2} + \alpha_{\text{O}_2}^k N_{\text{O}_2}]^{-1}$ is the total lifetime of k -state, α_X^k is a quenching rate of k due to collisions with molecule of type X of density N_X . The sum over the $n_m A_m$ represents the cascading from higher energy states. All the coefficients for the three bands are given in table 1 (Liu and Pasko 2004). It is important to note that in our work, as well as in Liu and Pasko (2004), equation (6) for excited states is solved with the streamer equations (1)–(4). This gives a full time-dependent solution of optical emissions in the modelling of the streamer processes. In equation (5), the radiosity $A_k n_k$ is an axially symmetric function $f(r)$. Then to calculate the intensity for a line of sight perpendicular to the discharge axis, it is necessary to transform equation (5); this is the classical direct Abel's transformation. As electrodes are in the computational domain, for lines of sight that cross the electrodes, it is necessary to modify the limits of Abel's transformation integral, as electrodes are opaque to the emission of the three bands studied. The integral obtained can be easily evaluated numerically using for example a classical Romberg method. Experimentally, the measured emission at a given time t is not instantaneous, but averaged during a time gate T_{gate} :

$$I_{T_{\text{gate}},k} = \frac{1}{T_{\text{gate}}} \int_{t-T_{\text{gate}}}^t I_k(t) dt. \quad (7)$$

3. Results and discussion

3.1. Influence of the data set used for transport parameters and source terms

In this section, we first simulate the streamer dynamics for an applied voltage of 7 kV and a pre-ionization level of

Table 1. Rate coefficients for optical emissions of 1PN_2 , 2PN_2 and 1NN_2^+ bands.

	$A_k \text{ (s}^{-1}\text{)}$	$\alpha_{\text{N}_2}^k \text{ (cm}^3 \text{ s}^{-1}\text{)}$	$\alpha_{\text{O}_2}^k \text{ (cm}^3 \text{ s}^{-1}\text{)}$
1NN_2^+	1.4×10^7	4×10^{-10}	
1PN_2	1.7×10^5	1×10^{-11}	
2PN_2	2.0×10^7		3×10^{-10}

10^9 cm^{-3} at $t = 0 \text{ s}$. This pre-ionization was chosen in Celestin *et al* (2009) as an estimation of the density of seed charges in the interelectrode gap for a repetitively pulsed discharge in the 10–30 kHz voltage frequency range. This test case is the same as the one used in Celestin *et al* (2009) except for two points. The first one is that in Celestin *et al* (2009) the air neutral density was $N_0 = 2.45 \times 10^{25} \text{ m}^{-3}$ at ground pressure and ambient temperature ($T_0 = 300 \text{ K}$) based on Kulikovskiy (1997). In this work, as in Bourdon *et al* (2007), we use $N_0 = 2.688 \times 10^{25} \text{ m}^{-3}$ for $T_0 = 273 \text{ K}$. This change has a negligible influence on the value of the density $N = N_0 T_0 / T$ at $T = 1000 \text{ K}$. The most significant change is on the data set used for transport parameters and reaction rates for air. In Celestin *et al* (2009), transport parameters and reaction rates were taken from Kulikovskiy (1997). In this work, they are taken from Morrow and Lowke (1997). For double-headed streamers propagating in high field, Liu and Pasko (2004) mentioned that the change in parameters may lead to deviations as large as 20% of streamer characteristics (i.e. electric field in the streamer head and body, streamer velocity, streamer radius, streamer electron density). In this section, we study the influence of the change in transport parameters and reaction rates on positive and negative streamers ignited at point electrodes and propagating in a weak field. As mentioned in the introduction, this study can be considered as a first step of a more detailed study on the influence of the change in the air composition on the discharge characteristics.

Figure 1 shows the time sequence of the distributions of the electron density and absolute values of the electric field. Figures 2 and 3 show the evolutions of axial profiles of the electric field and electron density, respectively. Due to the relatively high pre-ionization level, we have checked that photoionization has no influence on the results presented in this section. The dynamics of the discharge is essentially the same as the one obtained in Celestin *et al* (2009) with a positive streamer and a negative streamer propagating from the anode and cathode, respectively. After the interaction of both streamers, the positive streamer propagates very rapidly towards the cathode in the volume pre-ionized by the negative discharge. In comparison with the results obtained in Celestin *et al* (2009), with Morrow's coefficients, we note that the propagation of both discharges is faster and the interaction of both discharges occurs at about $t = 4 \text{ ns}$, that is to say about 1 ns earlier than in Celestin *et al* (2009). Before the discharge interaction, we note that the peak in the electric field in the streamer heads is higher with Kulikovskiy's coefficients. If we quantify the differences observed on streamer characteristics, we also obtain deviations as large as 20% for this point-to-point configuration. These results clearly show that the choice

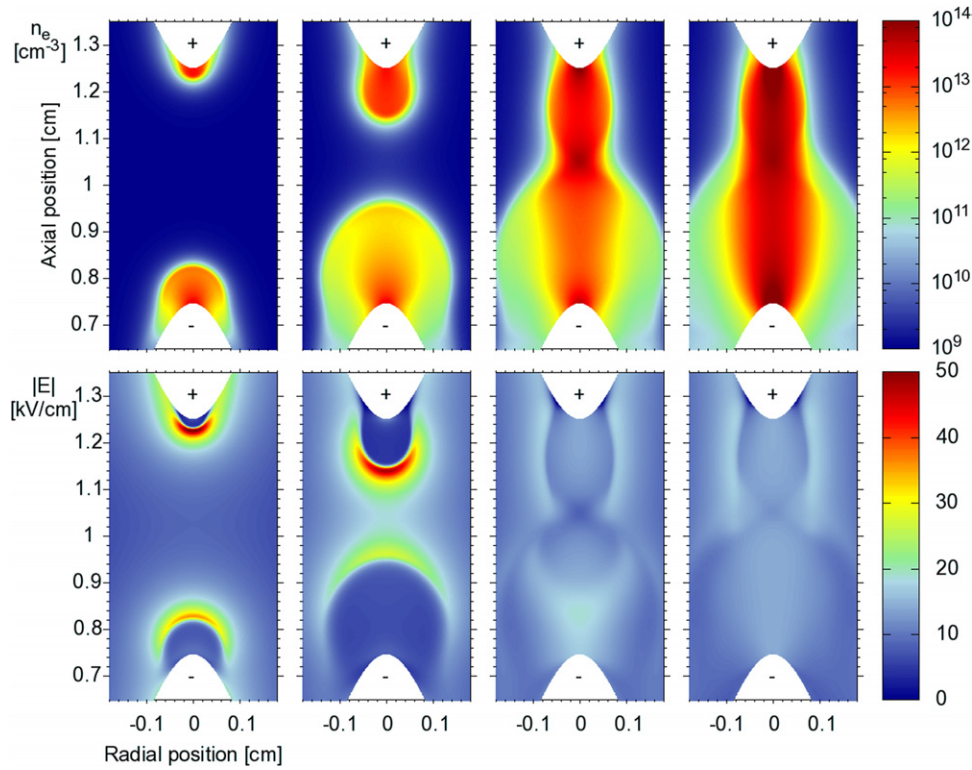


Figure 1. For a pre-ionization background of 10^9 cm^{-3} : a cross-sectional view of the electron density and the magnitude of the electric field at $t = 1, 3, 5$ and 7 ns .

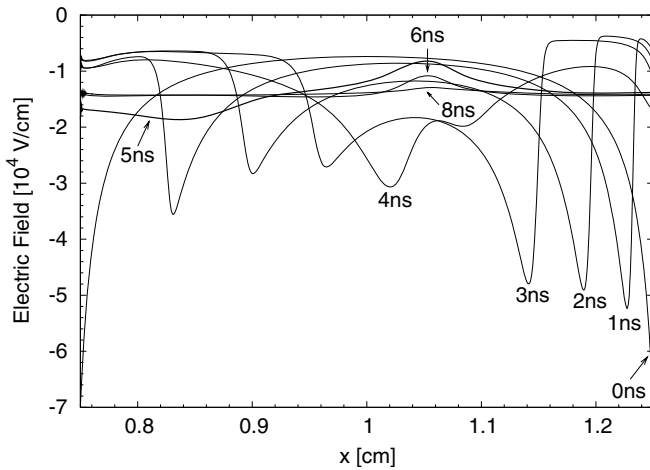


Figure 2. For a pre-ionization background of 10^9 cm^{-3} : electric field along the axis of symmetry at $t = 0\text{--}8 \text{ ns}$, with a time step of 1 ns .

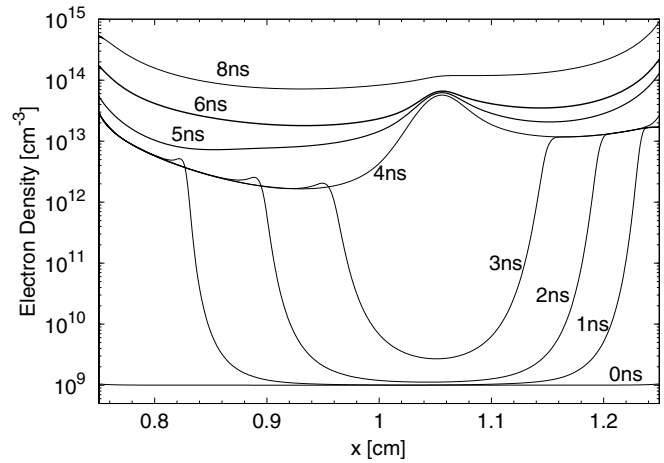


Figure 3. For a pre-ionization background of 10^9 cm^{-3} : electron density along the axis of symmetry at $t = 0\text{--}8 \text{ ns}$, with a time step of 1 ns .

of the transport parameters and reaction rates can have a non-negligible influence on the structure and dynamics of streamer discharges in air.

3.2. Influence of the pre-ionization background

In the previous section, we have considered that due to the high frequency of the applied voltage ($10\text{--}30 \text{ kHz}$), the pre-ionization background at $t = 0 \text{ s}$ was at a value of 10^9 cm^{-3} . In this section, we decrease the value of the pre-ionization background to study its influence on the discharge structure and dynamics. As already mentioned in the introduction, for

the studied discharge regime, a change in the pre-ionization level corresponds to a change in the applied voltage frequency. For a pre-ionization background of 10^9 cm^{-3} , we have checked that photoionization has no influence on the results. Then, in this section, we also study the relative importance of the pre-ionization background and of photoionization when the pre-ionization level is decreased. For atmospheric pressure and discharge repetition rates of $0.5\text{--}10 \text{ Hz}$, [Pancheshnyi \(2005\)](#) has estimated that there is still a residual charge accumulation between pulses at a level between 10^5 and 10^7 cm^{-3} . In this section, we consider the lowest limit of pre-ionization

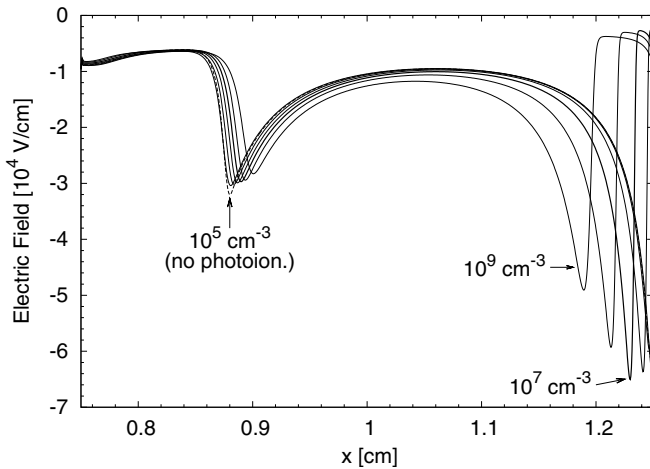


Figure 4. Electric field along the axis of symmetry at $t = 2$ ns for various pre-ionization backgrounds at $t = 0$. Solid lines: results are shown for pre-ionization background values from 10^4 to 10^9 cm^{-3} , with a multiplier step of 10. Dashed line: results obtained for a pre-ionization background of 10^5 cm^{-3} , without photoionization.

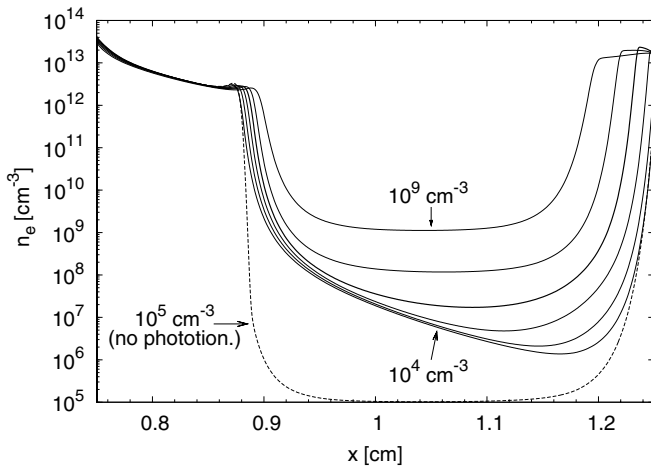


Figure 5. Electron density along the axis of symmetry at $t = 2$ ns for various pre-ionization backgrounds at $t = 0$. Solid lines: results are shown for pre-ionization background values from 10^4 to 10^9 cm^{-3} , with a multiplier step of 10. Dashed line: results obtained for a pre-ionization background of 10^5 cm^{-3} , without photoionization.

as 10^4 cm^{-3} , ten times less than the one proposed by Pancheshnyi (2005) in order to study a wide range of pre-ionization values from 10^4 and up to 10^9 cm^{-3} .

Figures 4 and 5 show the axial profiles of the electric field and electron density, respectively, at $t = 2$ ns for 6 values of the pre-ionization background. For a pre-ionization background of 10^5 cm^{-3} , figure 6 shows the time sequence of the distributions of the electron density and absolute values of the electric field and figures 7 and 8 show the evolutions of axial profiles of the electric field and electron density, respectively. In figures 4 and 5, we note that the significant decrease in the pre-ionization background from 10^9 to 10^4 cm^{-3} has a small influence on the electron density, electric field and location of the negative streamer head. Conversely, the decrease in the pre-ionization background has a significant

influence on the positive streamer characteristics. For a pre-ionization background of 10^9 cm^{-3} , the positive streamer starts to propagate immediately. Conversely, as the pre-ionization level decreases, figures 3 and 8 show that the delay for the positive streamer start, increases. As expected, the electric field and electron density in the positive streamer head decrease as the pre-ionization background increases. The comparison of figures 1 and 6 also shows that the radial expansion of both positive and negative streamers is slightly larger for a higher value of the pre-ionization level.

Figures 4 and 5 show that the level of charges in the interelectrode gap between the two discharges at $t = 2$ ns strongly depends on the pre-ionization level at $t = 0$ s. To clarify this point, in figures 4 and 5, we present also the results obtained for a pre-ionization level of 10^5 cm^{-3} without photoionization. In comparison with the case with photoionization, we note that the propagation of the negative streamer is only slightly slowed down. The most significant changes are observed ahead of the negative streamer front, on the level of seed charges and on the positive streamer side. When photoionization is taken into account, figure 6 shows that the positive streamer starts to propagate between 2 and 3 ns. Conversely, without photoionization, we have checked that the positive streamer starts only about ~ 1 ns later. Without photoionization, the level of seed charges in the interelectrode gap remains at the value of the pre-ionization background at $t = 0$ s. Thus, figures 4 and 5 clearly show that photoionization is not crucial for negative streamer propagation; however, the photoionization source term increases the number of seed charges ahead of the negative streamer front. For a pre-ionization background higher than 10^7 cm^{-3} , we note that both positive and negative streamers are propagating with a level of seed charges ahead of both streamer fronts which remains at the value of the pre-ionization background at $t = 0$ s. Conversely, for a pre-ionization background less than 10^7 cm^{-3} , the level of seed charges in the interelectrode gap increases in time. For this point-to-point configuration and a pre-ionization background less than 10^7 cm^{-3} , we note that seed electrons created by photoionization ahead of the negative streamer front contribute to the increase in seed charges in the interelectrode gap, and therefore to the propagation of the positive streamer head.

3.3. Optical emissions of the discharge

In this section, we present the optical emissions of the three bands 1PN_2 , 2PN_2 and 1NN_2^+ for the reference case considered in section 3.1 (and shown in figures 1–3) with a pre-ionization background of 10^9 cm^{-3} . Figure 9 presents the instantaneous emissions at $t = 3.5$ ns. At atmospheric pressure and ambient temperature, the lifetimes of the upper states $\text{N}_2(B^3\Pi_g)$, $\text{N}_2(C^3\Pi_u)$ and $\text{N}_2^+(B^2\Sigma_u^+)$ responsible for 1PN_2 , 2PN_2 and 1NN_2^+ are 5 ns, 0.6 ns and 0.1 ns, respectively. For air preheated at $T = 1000$ K, the lifetimes of $\text{N}_2(B^3\Pi_g)$, $\text{N}_2(C^3\Pi_u)$ and $\text{N}_2^+(B^2\Sigma_u^+)$ are 17 ns, 2 ns and 0.4 ns, respectively. The first negative system of N_2^+ has the shortest lifetime and the highest excitation threshold (~ 18.8 eV) therefore its radiation intensity is enhanced in the regions of high electric field, i.e. in the

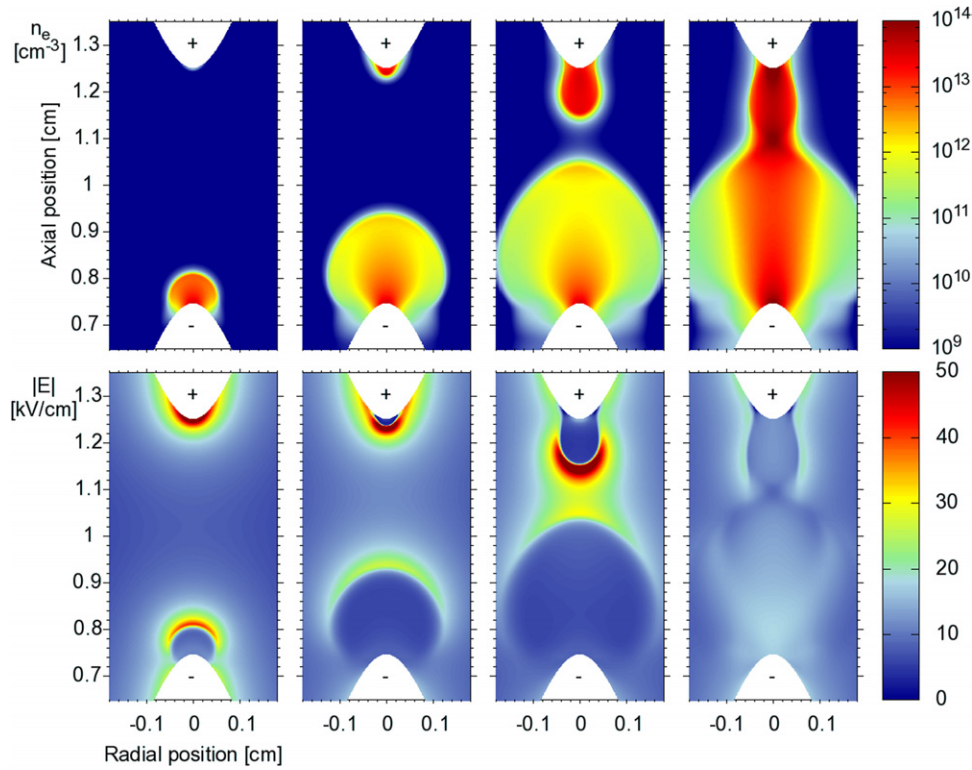


Figure 6. For a pre-ionization background of 10^5 cm^{-3} : a cross-sectional view of the electron density and the magnitude of the electric field at $t = 1, 3, 5$ and 7 ns .

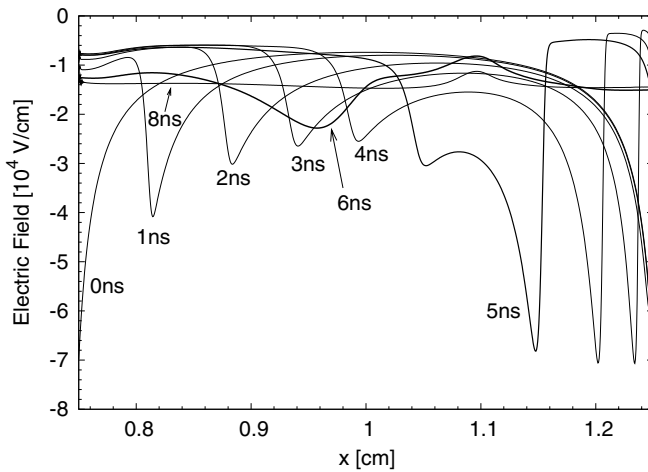


Figure 7. For a pre-ionization background of 10^5 cm^{-3} : electric field along the axis of symmetry at $t = 0\text{--}8 \text{ ns}$, with a time step of 1 ns .

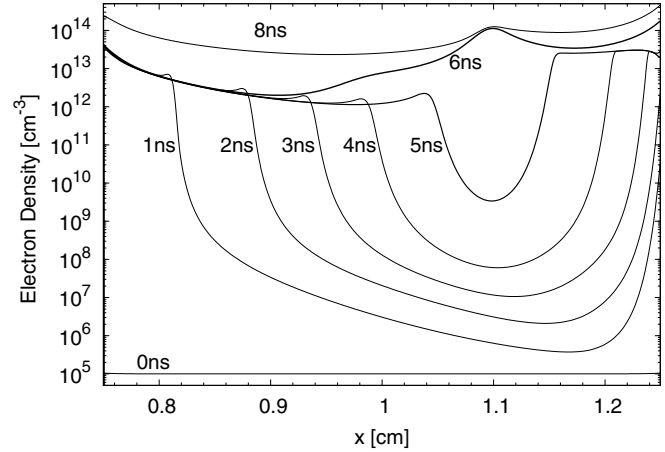


Figure 8. For a pre-ionization background of 10^5 cm^{-3} : electron density along the axis of symmetry at $t = 0\text{--}8 \text{ ns}$, with a time step of 1 ns .

streamer heads. The first positive bands of N_2 have the lowest excitation threshold ($\sim 7.35 \text{ eV}$) and are effectively excited by electrons in streamer heads and also in the relatively low field streamer bodies. As the effective lifetime of the 1PN_2 is 17 ns , on a timescale of the simulation presented in figure 9 (3.5 ns), the emission of the 1PN_2 band system forms a long trail of rather uniform intensity. The second positive bands of N_2 have an effective lifetime of 2 ns which is comparable to the timescale of the simulation presented in figure 9. The 2PN_2 excitation threshold is $\sim 11 \text{ eV}$ and then the emission of the 2PN_2 is enhanced in the high field regions of the streamers heads, but is also present in the relatively

low field regions of the streamer bodies. To illustrate the effect of temporal resolution, in figure 10 we present the average intensities of the three bands for a fixed ending time of $t = 3.5 \text{ ns}$ and a time interval of 2 ns , which corresponds to the time resolution of the Princeton Instrument PI-Max ICCD camera used for experimental studies (Pai 2008). In our simulations, the results are stored every 0.1 ns , which allows us to carry out an accurate time-averaging over 2 ns . As mentioned in Liu *et al* (2009), to understand the differences between instantaneous and time-averaged intensities of each band, two timescales are important: first, the lifetime of the upper states leading to the studied emission and second, a

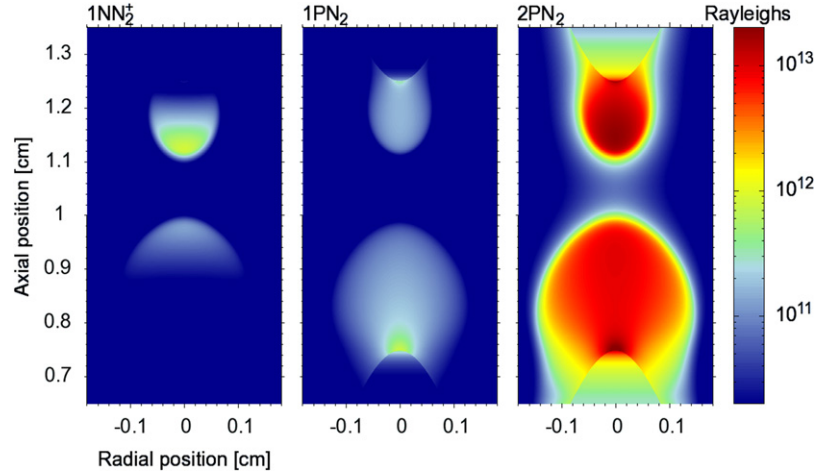


Figure 9. Instantaneous optical emissions of the three bands $1NN_2^+$, $1PN_2$ and $2PN_2$ at $t = 3.5$ ns for the same discharge conditions as in figure 1.

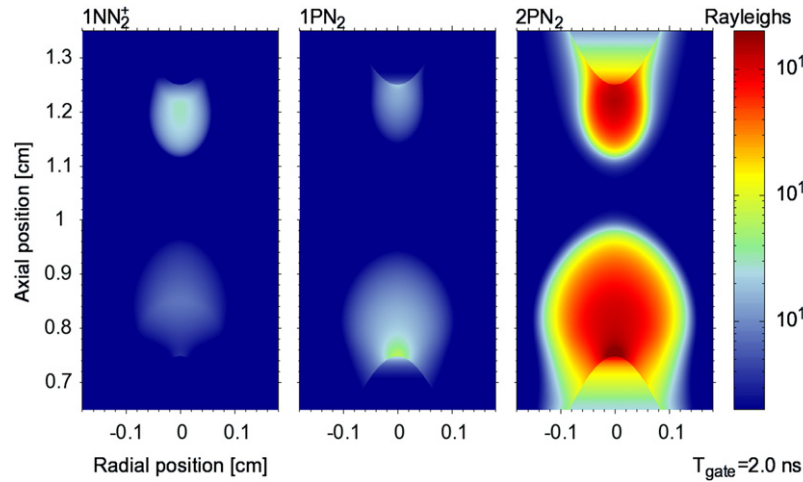


Figure 10. Time-averaged optical emissions of the three bands $1NN_2^+$, $1PN_2$ and $2PN_2$ at $t = 3.5$ ns with $T_{\text{gate}} = 2$ ns for the same discharge conditions as in figure 1.

characteristic timescale of the streamer propagation. For a given band, when the time interval for averaging is similar or less than the shortest of these two timescales, the result appears very similar to that of the instantaneous model output (Liu *et al* 2009). As already mentioned, for air at $T = 1000$ K, the lifetimes of $N_2(B^3\Pi_g)$, $N_2(C^3\Pi_u)$ and $N_2^+(B^2\Sigma_u^+)$ are 17 ns, 2 ns and 0.4 ns, respectively. Based on figures 2 and 3, we estimate the characteristic timescale of streamer propagation, to ~ 1 ns, which is similar to the time interval for averaging. In figure 10, we note that the time-averaged emissions of the $2PN_2$ and $1NN_2^+$ bands, which have the shortest lifetimes, become less structured and with a smaller radial expansion than in the instantaneous output shown in figure 9. Conversely, the structure of the time-averaged emission of the $1PN_2$, which has an effective lifetime of 17 ns, is almost unchanged but has a reduced intensity in comparison with the instantaneous result shown in figure 9. It is interesting to note that the time-averaged emission of the $2PN_2$ remains the strongest of the three bands, in agreement with experimental observations (Shcherbakov and Sigmond 2007, Pai 2008, Pancheshnyi *et al* 2005). Usually, different characteristics of

streamer discharges (e.g. streamer velocity, streamer radius) are derived from the measured $2PN_2$ intensity distributions. In this work, we show that, even with a short 2 ns time averaging, the emission of the $2PN_2$ is different from the instantaneous emission, and seems to indicate that both streamers have a smaller radial expansion and more diffuse streamer heads.

Finally, it is interesting to note that in (Pai 2008), the spatial resolution of optical emission measurements is $25\ \mu\text{m}$. In the region where the streamer propagates, the grid is Cartesian, with a fixed cell size of $5\ \mu\text{m}$. Then for a spatial resolution of $25\ \mu\text{m}$, we have calculated the intensity at each grid point for the reduced resolution image by averaging over 5×5 cells the intensities of the original grid points with the high-resolution image. The results obtained for a spatial resolution of $25\ \mu\text{m}$ are not shown here as they are very similar to instantaneous results. Then, for optical emission diagnostics in laboratory experiments, this work clearly shows that the most critical point for the analysis of measured optical emissions with currently available cameras is the temporal resolution and not the spatial resolution. It is interesting to note that for sprite observations, both temporal

and spatial resolutions of used instruments have to be taken into account (Liu *et al* 2009).

4. Conclusions

In this paper, we present simulations of the structure and the dynamics of positive and negative streamers propagating in preheated air at atmospheric pressure between two point electrodes. To compare with experiments carried out with a repetitively pulsed voltage, we have considered that many discharges have occurred before the simulated one, and then we have estimated that the density of seed charges in the interelectrode gap is on the order of 10^9 cm^{-3} for the 10–30 kHz voltage frequency range. In Celestin *et al* (2009), we have found that for this high value of pre-ionization background, the photoionization has a negligible influence on the results.

In this work, first we have studied the influence of the data set used for transport parameters and reaction rates for air on the simulation results. We have compared results obtained using the input parameters from Morrow and Lowke (1997) and those from Kulikovsky (1997) used in Celestin *et al* (2009). Deviations as large as 20% of streamer characteristics (i.e. electric field in the streamer head and body, streamer velocity, streamer radius, streamer electron density) have been observed for this weak field point-to-point configuration. For double-headed streamers propagating in high field, Liu and Pasko (2004) obtained also deviations as large as 20% of streamer characteristics. These results clearly show that the choice of transport parameters and reaction rates for air has a non-negligible influence on the discharge structure and dynamics. We consider that the use of different input data for transport parameters and reaction rates could explain differences observed between different streamer simulation results and may even impact the appearance or not of streamer branching in simulations. In this work, we have compared the results obtained using two data sets for transport parameters and reaction rates to make a first attempt before estimating the influence on the discharge of the change in the air composition between ambient temperature and $T = 1000 \text{ K}$. In a future work, it would be interesting to take into account real changes in the composition of air and its chemistry on the discharge structure and dynamics.

Second, we have studied the influence of the pulsed voltage frequency on the discharge structure. For the studied discharge regime, a change in the applied voltage frequency corresponds to a change in the pre-ionization background. We have considered a wide range of pre-ionization values from 10^4 up to 10^9 cm^{-3} . We have noted that a decrease in the pre-ionization background has a small influence on the electron density, electric field and location of the negative streamer head. Conversely, a decrease in the pre-ionization background has a significant influence on the positive streamer characteristics. For this test case, simulation results for low pre-ionization background show that seed electrons created by photoionization ahead of the negative streamer front contribute to the increase in seed charges in the interelectrode gap and therefore, to the propagation of the positive streamer head. It

is important to note that in this work we have used a simple boundary condition for charged species at the cathode. If secondary emission processes are included, preliminary results show that the discharge propagating from the cathode point is slightly weaker (i.e. lower peak electric field and electron density) than the negative streamer calculated in this work, which is always much weaker than the positive discharge. Then, we consider that the main conclusions of this work on the positive and negative discharge interaction are not affected by the change in boundary conditions at the cathode. However, the influence of boundary conditions at the cathode on the point-to-point discharge in preheated air will be presented in a separate dedicated paper.

Finally, we present the optical emissions of the discharge for the three bands 1PN_2 , 2PN_2 and 1NN_2^+ . For the reference test case with a pre-ionization background of 10^9 cm^{-3} , we compare instantaneous and time-averaged emissions of the three bands. The time averaging is carried out on a time interval of 2 ns, which corresponds to the time resolution of the camera used for experimental studies (Pai 2008). In instantaneous and time-averaged results, the emission of the 2PN_2 is the strongest of the three bands, in agreement with experimental observations. It is interesting to note that for a short 2 ns time interval, time-averaged emissions of the 2PN_2 and 1NN_2^+ are slightly different from instantaneous ones whereas the structure of the emission of the 1PN_2 is almost unchanged. Usually in the experiments, different characteristics of streamer discharges (e.g. streamer velocity, streamer radius) are derived from the 2PN_2 intensity recordings. In this work, it is interesting to note that the time-averaged emission of the 2PN_2 seems to indicate that both streamers have a smaller radial expansion and more diffuse streamer heads than those derived from instantaneous results. The detailed comparison of calculated and measured optical emissions of the discharge in the point-to-point geometry will be presented in a separate dedicated paper.

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